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Assignment - 3 Basic of ML

Que-1) what is Hypothis in machine learning?

- A hypothesis is a function that maps input features to output prediction. It represents the models assumptions about the Data and is adjusted during training to minimize the errors.

Que-2) Explain the impact of taking the learning rate? when can you take smaller value of learning rate and in which situation the value can be larger?

- The learning rate is a hyperparameter that controls how much the model's parameters are updated during each iteration of training.

- Smaller Learning rate: Leads to slow convergence but ensures stability and better accuracy. suitable when data is complex or noisy.

- Larger Learning rate: speeds up convergence but may the optimal point, risking divergence suitable for a simpler datasets.

- Choosing an appropriate learning rate is essential to ensure fast, stable and accurate training of a ML model.

Que-3) what is overfitting, and How can you Avoid it?

- overfitting accurs when a model Performs well on training data a but Poorly on unseen data.

- Prevention Techniques:

- cross validation
- Pruning (for decision trees)
- Regularization (L1, L2)
- Early stopping
- Increasing training Data
- Reducing model complexity

Que-4) why the use of linear regression is not Preferred for a classification task?

- Linear regression Predicts continuos values and lacks clear decision boudaries, making it ineffective for discrete class labels.

- logistic regression or decision trees are better suited for classification.

Que-5) why do we Perform normalization?

- Normalization scales features to a common range (e.g. 0-1) improving model convergence speed and accuracy. It's crucial for Distance based algorithms like KNN.

K-means and gradient descent.

Que-6) what is the bias-variance tradeoff?

- The bias-variance tradeoff refers to the balance between two types of errors that affect a machine learning model's Performance.

- High Bias: underfitting model is too simple
- High variance: overfitting model is too complex
- The tradeoff aims to find a balance for optimal Performance.

Que-7) what is 'training set' and 'test set' in machine learning model? what is the requirement of training validation, and Test sets?

- Traing set: used to train the model.
- Test set: used to evaluate model Performance
- validation set: Helps tune hyperparameters to improve Performance.

Que-8) How do you Handle missing or corrupted Data in a Dataset?

- Imputation: filling missing value using mean, median or mode.
- Deletion: Remove rows or columns with excessive missing Data.
- Prediction models: use models to Predict missing value.

- Flagging: Add indicators for missing values.

Que-9) Explain the evaluation matrix for classification Problem.

- Confusion matrix: A table used to describe the performance of classification model by comparing actual and predicts value.

- True Positive (TP): model correctly predicts the positive class.
- True negative (TN): model correctly predicts the negative class.
- False Positive (FP): model predicts positive but the actual class is negative.
- False negative (FN): model predicts negative but the actual class is positive.

- Accuracy: measures overall correctness of the model.
- Precision: measures how many predicted positive class are correctly predicted.
- Recall: measures how many predicted positive cases are actually positive.
- F1 score: Harmonic mean of precision and recall, useful for imbalanced datasets.

Que-10) Explain the evaluation matrix for Regression Problem.

- MAE (mean Abousolute Error): Average absolute error between actual and predict values.

- MSE (mean squared error): Average of squared error.

Penalizes large errors.

- RMSE (Root mean squared error): Square root of MSE.

easy to interpet.

- R2 score: measures how well the model explains data variance.